

VISHNU ASUTOSH DASU

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EDUCATION

- **Pennsylvania State University** *Aug 2024 - May 2028*
Doctor of Philosophy, Computer Science and Engineering
 - *Research Area:* Trustworthy AI, LLMs for Code, Program Analysis, and Software Security
 - *Advisor:* Prof. Gang (Gary) Tan
- **Pennsylvania State University** *Aug 2022 - May 2024*
Master of Science, Computer Science and Engineering
 - *Thesis:* “Mitigating Unfairness in Deep Learning”
 - *Advisor:* Prof. Gang (Gary) Tan
- **Manipal Institute of Technology (MIT), Manipal** *July 2016 - July 2020*
Bachelor of Technology, Computer Science and Engineering

RESEARCH INTERESTS

I am a trustworthy AI and software security researcher. I build automated, scalable pipelines that discover failure modes in LLMs and software—unfair, non-robust, privacy-leaking, or incorrect behavior—and turn them into targeted repair, data curation, and training signals. My work combines machine learning, search-based optimization, and program analysis to improve the coding capabilities, correctness, fairness, robustness, and privacy of AI and software systems.

Keywords: LLMs for code, trustworthy AI, failure-driven repair and data curation, program analysis, privacy-preserving machine learning

SELECTED PUBLICATIONS

* - Equal Contribution, † - Alphabetical Order

- **Attention Pruning: Automated Fairness Repair of Language Models via Surrogate Simulated Annealing** [pdf]
48th International Conference on Software Engineering (ICSE), 2026
Vishnu Asutosh Dasu, Md Rafi ur Rashid, Vipul Gupta, Saeid Tizpaz-Niari, Gang Tan
- **Improving Noise Efficiency in Privacy-preserving Dataset Distillation** [pdf]
International Conference on Computer Vision (ICCV), 2025
Runkai Zheng, Vishnu Asutosh Dasu, Yinong Wang, Haohan Wang, Fernando De la Torre
- **Privacy-Preserving Data Deduplication for Enhancing Federated Learning of Language Models** [pdf]
Network and Distributed System Security (NDSS) Symposium, 2025
Aydin Abadi*, Vishnu Asutosh Dasu*, Sumanta Sarkar*
- **Chain-of-Thought Driven Adversarial Scenario Extrapolation for Robust Language Models** [pdf]
40th Annual AAAI Conference on Artificial Intelligence (AAAI), 2026
Md Rafi Ur Rashid, Vishnu Asutosh Dasu, Ye Wang, Gang Tan, Shagufta Mehnaz
- **NeuFair: Neural Network Fairness Repair with Dropout** [pdf]
33rd ACM SIGSOFT International Symposium on Software Testing and Analysis (ISSTA), 2024
Vishnu Asutosh Dasu, Ashish Kumar, Saeid Tizpaz-Niari, Gang Tan

- **Gradient-Free Privacy Leakage in Federated Language Models through Selective Weight Tampering [pdf]**
Privacy Enhancing Technologies Symposium (PETS) 2026
 Md Rafi Ur Rashid, **Vishnu Asutosh Dasu**, Kang Gu, Najrin Sultana, Shagufta Mehnaz
- **PROV-FL: Privacy-preserving Round Optimal Verifiable Federated Learning [pdf]**
15th ACM Workshop on Artificial Intelligence and Security (AISEC), ACM CCS, 2022
Vishnu Asutosh Dasu, Sumanta Sarkar, Kalikinkar Mandal
- **Side Channel Attack On Stream Ciphers: A Three-Step Approach To State/Key Recovery [pdf]**
IACR Conference on Cryptographic Hardware and Embedded Systems (CHES), 2022
 Satyam Kumar, **Vishnu Asutosh Dasu**, Anubhab Baksi, Santanu Sarkar, Dirmanto Jap, Jakub Breier, Shivam Bhasin
- **Impact of Data Duplication on Deep Neural Network-Based Image Classifiers: Robust vs. Standard Models [pdf]**
IEEE Deep Learning Security and Privacy Workshop (DLSP), IEEE S&P, 2025
 Alireza Aghabagherloo, Aydin Abadi, Sumanta Sarkar, **Vishnu Asutosh Dasu**, Bart Preneel
- **EvoquerBot: A multimedia chatbot leveraging synthetic data for cross-domain assistance [pdf]**
Alexa Prize TaskBot Challenge 2 Proceedings
 Team EvoquerBOT, Penn State University
- **New Results on Machine Learning-Based Distinguishers [pdf]**
IEEE Access, 2023
 Anubhab Baksi[†], Jakub Breier[†], **Vishnu Asutosh Dasu**[†], Xiaolu Hou[†], Hyunji Kim[†], Hwajeong Seo[†]
- **Three Input Exclusive-OR Gate Support For Boyar-Peralta's Algorithm [pdf]**
22nd International Conference on Cryptology in India (Indocrypt), 2021
 Anubhab Baksi, **Vishnu Asutosh Dasu**, Banashri Karmakar, Anupam Chattopadhyay, Takanori Isobe
- **LIGHTER-R: Optimized Reversible Circuit Implementation For SBoxes**
32nd IEEE International System-on-Chip Conference (SOCC), 2019
Vishnu Asutosh Dasu, Anubhab Baksi, Sumanta Sarkar, Anupam Chattopadhyay
- **[Re] GANSpace: Discovering Interpretable GAN Controls**
ReScience C, 2022
Vishnu Asutosh Dasu, Midhush Manohar T.K.

PREPRINTS/UNDER REVIEW

- **Heimdall: Formally Verified Automated Migration of Legacy eBPF Programs to Rust [pdf]**
(Under Review) USENIX Security 2027
Vishnu Asutosh Dasu, Monika Santra, Md Rafi Ur Rashid, Ashish Kumar, Saeid Tizpaz-Niari, Gang Tan
- **Robustness of Vision Language Models Against Split-Image Harmful Input Attacks [pdf]**
(Minor Revision) ACM Conference on Computer and Communications Security (CCS) 2026
 Md Rafi Ur Rashid, Md Sadik Hossain Shanto, **Vishnu Asutosh Dasu**, Shagufta Mehnaz
- **iResolveX: Multi-Layered Indirect Call Resolution via Static Reasoning and Learning-Augmented Refinement [pdf]**
(Under Review) Network and Distributed System Security (NDSS) Symposium, 2027
 Monika Santra, Bokai Zhang, Mark Lim, **Vishnu Asutosh Dasu**, Dongrui Zeng, Gang Tan

- **Finding Probabilistic Sub-invariants via Search Space Algorithms under Noisy Cost Functions**
(Under Review) *International Workshop on Formal Methods for Autonomous Systems (FMAS) 2026*
Ashish Kumar, Vishnu Asutosh Dasu, Saeid Tizpaz-Niari, Gang Tan

PATENTS

- **A Method to Secure Dynamic Agentic Artificial Intelligence Systems from Prompt Injection Attacks**
(Filing in Progress) 2026
Vishnu Asutosh Dasu, Lilit Avetisyan, Jorge Guajardo Merchan

ACADEMIC AND WORK EXPERIENCE

- **Bosch Center for AI** May 2026 - Aug. 2026
AI Research Intern Advisor: Jorge Guajardo Merchan
– Projects: System-level Defenses for Securing AI Agents
- **Tata Research Development and Design Centre (TRDDC)** Sept 2020 - June 2022
Researcher, Cybersecurity and Privacy Advisor: Sumanta Sarkar
– Projects: Privacy-preserving Federated Learning, Secure Multiparty Computation, Insider Threat Detection
- **Citrix R&D** Jan 2020 - June 2020
Software Engineer Intern, Citrix Analytics for Security (CAS)
– Full-stack Web Developer (Spring Boot, React.js)
- **Nanyang Technological University (NTU)** Dec 2019 - Jan 2020
Research Intern Advisor: Anupam Chattopadhyay
– Projects: Optimizing hardware implementations of block ciphers
- **Tata Consultancy Services (TCS) Research** May 2019 - July 2019
Research Intern, Cybersecurity and Privacy Advisor: Chalamala Srinivasa Rao
– Projects: Adversarial attacks and defenses on Convolutional Neural Networks (CNNs)
- **Indian Statistical Institute** May 2018 - July 2018
Summer Scholar Advisor: Dipti Prasad Mukherjee
– Project: 3-D coordinate estimation of humans from 2-D live video feed

SKILLS

- **Languages:** Python, C/C++, Rust, Java, Go, SQL
- **AI/ML:** LLMs, VLMs, and Code LLMs (pre-training, alignment, and post-training), synthetic data generation, data curation and deduplication, quantization, agentic pipelines, Trustworthy AI (Fairness, Privacy, Robustness)
- **Frameworks:** PyTorch, Hugging Face Transformers, PEFT, NumPy, Pandas, Scikit-learn
- **Program Analysis & Verification:** LLVM, Clang, eBPF, Z3/SMT solvers, KLEE, angr, symbolic execution
- **Areas of Expertise:** Trustworthy AI, LLMs for Code, Software Security, Program Analysis

SERVICE

- **Program Committee:** USENIX Security 2027, AAAI 2027 (Senior Program Committee), AAAI 2026

- **Reviewer:** IEEE TDSC (2026), IEEE TIFS (2025, 2026), IEEE Access (2025)
- **Artifact Evaluation Committee:** ASE 2026, ISSTA 2026, ACM CCS 2026, USENIX Security 2026, ACM CCS 2025
- **External Reviewer:** NDSS 2025, OOPSLA 2025, IEEE S&P 2025, IEEE S&P 2026
- **Organizer:** Penn State Security Reading Group (2024 - Present)
- **Judge:** Penn State Undergraduate Exhibition 2025

AWARDS AND ACHIEVEMENTS

- **Distinguished Reviewer Award:** Received the distinguished reviewer award for artifact evaluation in USENIX Security 2026.
- **Future Leader of AI:** Selected as a Future Leader of AI at the inaugural *ACM AI Leadership Summit*, 2026 with a scholarship.
- **Scholarship:** Received a scholarship to attend the *Fifteenth Summer School on Formal Techniques* in Menlo Park, CA.
- **Student Travel Grant:** Received the ACM/SIGSOFT CAPS travel grant to attend ICSE 2026.
- **Distinguished Reviewer Award:** Received the distinguished reviewer award for artifact evaluation in ACM CCS 2025.
- **Student Travel Grant:** Received the travel grant to attend ICCV 2025.
- **Internet Society Fellowship:** Received the Internet Society Fellowship to attend NDSS 2025.
- **TCS Citation Award ($3 \times$ recipient):** Received the award for outstanding research contributions to Tata Consultancy Services (TCS).
- **Scholarship:** Received a scholarship to attend the *Winter School on Responsible AI* in Israel.
- **Best Project Award:** Received the Best Project Award during the *Fifth Summer School on Computer Vision, Graphics and Image Processing*, Indian Statistical Institute (ISI) Kolkata.
- **IGVC:** Placed 2nd in the Interoperability Profiles Challenge and 9th overall at *Intelligent Ground Vehicle Competition (IGVC)* 2018. Second-best among all teams from India.
- **ACM ICPC Regionals:** Represented MIT Manipal at the 2017 *ACM ICPC Asia Regional Contest*.
- **DAGsHub Award:** Received a \$500 award from *DAGsHub* for completing the *ML Reproducibility Challenge Spring 2021*.

GRANTS

- **NSF NAIRR 260079:** Automated and Verified Migration of Legacy eBPF Programs to Rust. Total Award: \$8,000. 2026–2027.

PROFESSIONAL MEMBERSHIPS

- Student Member of the ACM
- Student Member of IEEE

REFERENCES

- Prof. Gang Tan (Advisor): gtan@psu.edu
- Prof. Saeid Tizpaz-Niari: saeid@uic.edu
- Prof. Sumanta Sarkar: sumanta.sarkar@essex.ac.uk